

Functional Features included in the solution

Date	26 June 2026
Team ID	SWTID-2026-9014
Project Name	AdForge AI: AI-Powered Advertisement Copy Generator
Maximum Marks	5 Marks

Functional Features Included in the Solution:

AdForge AI is a Generative AI-powered Advertisement Copy Generation Platform built using Python Flask and the Qwen3-32B language model via the Groq API. The application enables marketers, startups, creators, and business owners to generate platform-specific, conversion-optimised advertisement copy for Google Ads, Instagram, and Facebook within seconds. The solution comprises three core AI-powered modules — the Ad Generator, the Advertisement Refinement Engine, and the Performance Intelligence Generator — supported by a secure Flask backend, a responsive premium SaaS frontend, a content safety filter, an automated test suite of 60 tests, and a public deployment pipeline via Ngrok. All 20 functional features listed below have been fully implemented and successfully demonstrated in the final project submission.

S · N o	Feature Name	Description & Purpose	Module / Layer	Status
1	Multi-Platform Ad Generator	Accepts a complete campaign brief from the user and generates platform-specific advertisement copy simultaneously for Google Ads, Instagram, and Facebook in a single AI request. Each platform receives uniquely structured output tailored to its	Ad Generator Module — Backend app.py + Frontend index.html	Complete

		format, character limits, and audience behaviour. Helps marketers produce ready-to-publish ads across all major platforms within seconds.		
2	Google Ads Copy Generation	Generates 15 unique headlines each within the 30-character Google Ads limit and 4 keyword-optimised descriptions each within the 90-character limit. All headlines are non-repetitive, high CTR focused, and include the user's keywords and call to action as required by Google's responsive search ad format.	Ad Generator Module — build_ad_prompt() in app.py	Completed
3	Instagram Ads Copy Generation	Generates 3 unique caption variations for Instagram using storytelling hooks, intelligent emoji usage, engagement questions, and brand voice. Each caption is distinct in angle — one storytelling, one value-driven, and one FOMO-based. Output includes 20 to 25 targeted hashtags and a clear call to action optimised for Instagram's algorithm and audience engagement patterns.	Ad Generator Module — build_ad_prompt() in app.py	Completed

4	Facebook Ads Copy Generation	Generates 3 primary text variations for Facebook using conversational style, social proof, emotional triggers, and value-driven messaging. Each variation includes a unique ad headline, description, and call to action. Output is structured to match Facebook's ad format requirements and optimised for the news feed placement.	Ad Generator Module — build_ad_prompt() in app.py	Completed
5	Marketing Slogan Generator	Automatically generates 5 unique, memorable, and brandable marketing slogans tailored to the product, USP, industry, and target audience as part of every ad generation request. Each slogan is kept under 10 words for maximum impact and brand recall. Helps businesses establish a consistent brand voice across all marketing channels.	Ad Generator Module — build_ad_prompt() in app.py	Completed
6	16 Marketing Tone Options	Provides a dropdown selection of 16 distinct marketing tones — Professional, Playful,	Ad Generator Module — index.html tone dropdown + build_ad_prompt()	Completed

		Luxury, Minimal, Emotional, Bold, Confident, Friendly, Urgent, Persuasive, Inspirational, Casual, Corporate, Youthful, Creative, and Premium. The selected tone is passed into the AI prompt and applied consistently across all generated content, ensuring the output matches the brand's personality and campaign objectives.		
7	Marketing Psychology Frameworks	The AI prompt instructs the Qwen3-32B model to reason internally and apply relevant marketing psychology techniques before generating copy. Frameworks applied include AIDA, PAS, FAB, BAB, Storytelling, Scarcity, Social Proof, Loss Aversion, FOMO, Emotional Selling, Authority, Reciprocity, and Urgency. Only techniques that naturally fit the campaign brief are applied, ensuring copy feels genuine	AI Prompt Engineering — build_ad_prompt() in app.py	Completed

		rather than formulaic.		
8	Advertisement Refinement Engine	Accepts an existing advertisement and specific user feedback instructions such as make it shorter, add urgency, target teenagers, use emotional marketing, or improve the CTA. Returns a fully improved version of the advertisement along with a detailed list of changes made with reasoning, an improvement score out of 10, and a conversion prediction. Eliminates the need to manually rewrite advertisements from scratch when adjustments are needed.	Refinement Module — refine_ad() route + build_refine_prompt() in app.py	Completed
9	Performance Intelligence Generator	Generates a complete Performance Intelligence Report for any campaign including 4 A/B testing ideas with hypotheses, 3 CTA improvement suggestions, SEO keyword recommendations, 3 conversion	Performance Tips Module — performance_tips() route + build_tips_prompt() in app.py	Completed

		optimisation tips, platform-specific best posting times, and 2 to 3 refined audience segment recommendations. Provides actionable marketing intelligence without requiring a marketing analyst.		
10	Content Safety Filter	Automatically screens every user input for blocked terms related to illegal products, fraud, adult content, violence, hate speech, scams, and misinformation before the request reaches the Groq API. Returns a polite 422 rejection response if harmful content is detected. Ensures the application cannot be misused to generate unethical or illegal marketing content under any circumstances.	Security Layer — <code>is_unsafe_input()</code> in <code>app.py</code>	Completed
11	Input Validation	Validates all required form fields on the server side before processing any request. Returns a descriptive 400 error response identifying	Validation Layer — <code>validate_inputs()</code> in <code>app.py</code>	Completed

		exactly which required field is missing. Separate validation rule sets are applied for the Ad Generator form and the Performance Tips form since they have different required fields. Prevents incomplete or malformed requests from reaching the AI model.		
12	Input Sanitisation	Every user input field is passed through the Bleach library which strips all HTML tags and enforces per-field character length limits before the data is used in any prompt or response. Prevents Cross-Site Scripting attacks, HTML injection, and prompt injection attempts. Ensures all data reaching the AI model is clean, safe, and within defined boundaries.	Security Layer — <code>sanitize()</code> using <code>bleach.clean()</code> in <code>app.py</code>	Completed
13	Asynchronous Form Submission	All three forms submit data asynchronously using the JavaScript Fetch API without any page reload. A	Frontend Layer — <code>script.js</code> <code>submitAdForm()</code> · <code>submitRefineForm()</code> · <code>submitTipsForm()</code>	Completed

		loading spinner with cycling messages appears while the AI processes the request. The submit button is disabled during the request to prevent duplicate submissions and re-enabled automatically when the response is received. Provides a smooth, modern SaaS-like user experience.		
14	Markdown Response Rendering	The AI returns all generated content in structured Markdown format. A custom lightweight JavaScript Markdown renderer parses the response and converts it to formatted HTML with headings, bold text, bullet lists, and horizontal rules — displayed inline in the output panel without any page reload. No external Markdown library is required.	Frontend Layer — renderMarkdown() in script.js	Completed
15	Copy to Clipboard	Every output panel includes a Copy button that instantly copies the full generated	Frontend Layer — copyOutput() + showToast() in script.js	Completed

		advertisement text to the user's system clipboard. A toast notification appears at the bottom right of the screen confirming the action. Allows users to immediately paste generated content into their advertising platforms, documents, or team chats.		
16	Download Output as TXT	Every output panel includes a Download TXT button that saves the generated advertisement copy as a plain text file to the user's device. Files are named descriptively as adforge-ads.txt, adforge-refined.txt, or adforge-tips.txt depending on the module. Allows users to archive, share, or present generated content outside the application.	Frontend Layer — downloadOutput() in script.js	Completed
17	About Page	A fully designed informational page accessible at /about covering the application purpose, all core modules, complete technology stack, security	Presentation Layer — templates/about.html + /about route in app.py	Completed

		features, and project context. Accessible from the navigation bar on all pages. Provides evaluators, users, and developers with a complete overview of what AdForge AI is and how it was built.		
18	How It Works Page	A fully designed technical walkthrough page accessible at /how-it-works covering the 7-step user journey, complete request data flow, application architecture, marketing psychology frameworks, local deployment guide, Ngrok public deployment instructions, and automated test results. Accessible from the navigation bar on all pages.	Presentation Layer — templates/how_it_works.html + /how-it-works route in app.py	Completed
19	Public Deployment via Ngrok	A dedicated ngrok_start.py script creates a secure public HTTPS tunnel to the locally running Flask application. Generates a live public URL of the format https://abc123.ngrok-	Deployment Layer — ngrok_start.py using pyngrok library	Completed

		free.app that is accessible from any device or browser worldwide without requiring any cloud hosting or server infrastructure. Ideal for demonstrations, client reviews, and academic submissions.		
20	Automated Test Suite	A comprehensive test_app.py file containing 60 automated tests across 6 categories — content safety filter, input sanitisation, prompt engineering, input validation, Flask route verification, and page content verification. All tests run without requiring a live Groq API key. Produces a clear pass/fail summary in the terminal. Ensures all core application logic is verified and regression-free after any code change.	Testing Layer — test_app.py	Completed